

Feature-Importance Report

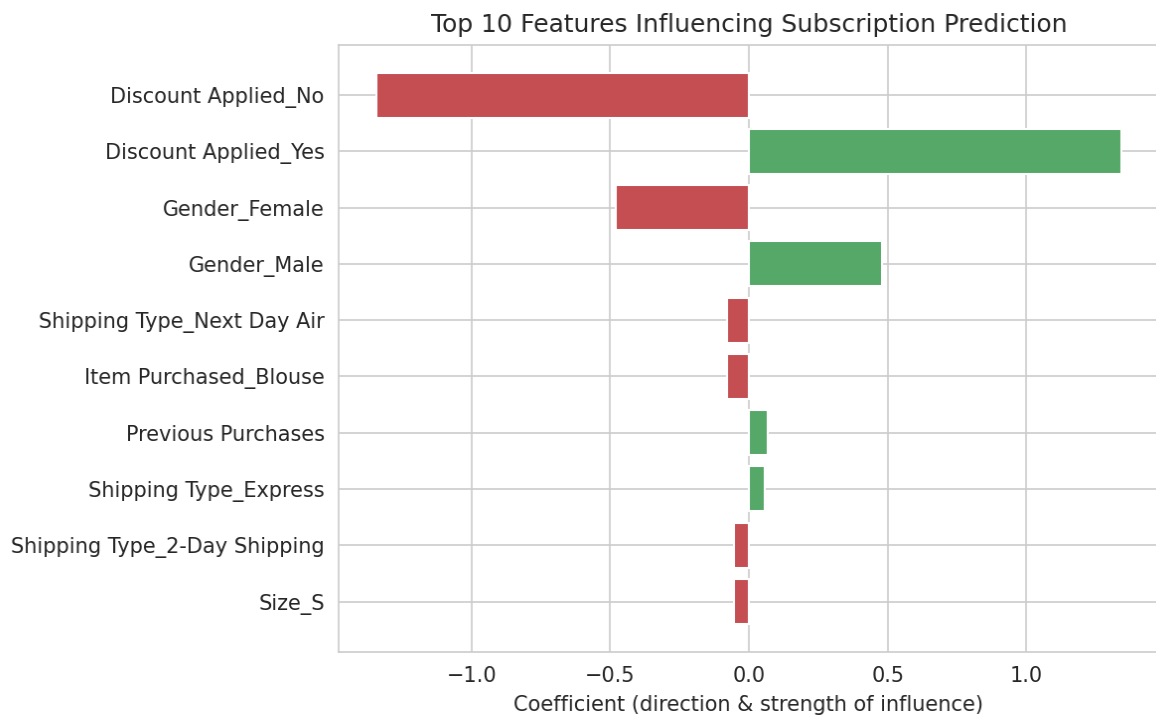
Task 9 — Optimized Classification Model for Subscription-Likelihood Prediction

1. Model and Method

The final selected model is an optimized Logistic Regression pipeline (preprocessing + classifier), tuned with 5-fold GridSearchCV optimizing F1-score. Because Logistic Regression is a linear model, feature importance is read directly from its coefficients: a positive coefficient increases the predicted probability of subscription, a negative coefficient decreases it, and the magnitude reflects the strength of that relationship (after all inputs have been standardized/encoded, so coefficients are comparable to one another).

2. Top 10 Influential Features

Rank	Feature	Coefficient	Direction
1	Discount Applied = No	-1.3434	Decreases likelihood
2	Discount Applied = Yes	+1.3425	Increases likelihood
3	Gender = Female	-0.4801	Decreases likelihood
4	Gender = Male	+0.4792	Increases likelihood
5	Shipping Type = Next Day Air	-0.0796	Decreases likelihood
6	Item Purchased = Blouse	-0.0790	Decreases likelihood
7	Previous Purchases (count)	+0.0668	Increases likelihood
8	Shipping Type = Express	+0.0585	Increases likelihood
9	Shipping Type = 2-Day Shipping	-0.0543	Decreases likelihood
10	Size = S	-0.0523	Decreases likelihood



3. Business Interpretation

Feature	Interpretation	Recommended Action
Discount Applied	By far the strongest signal. Customers who received a discount are much more likely to be subscribed, consistent with subscription programs commonly bundling discount perks.	Bundle a modest discount into subscription sign-up offers; test discount-led win-back offers for high-probability non-subscribers.
Gender	Second strongest signal, but every subscribed customer in this dataset happens to be male — this looks like a data-generation artifact rather than a real driver of loyalty.	Do not use gender as a targeting criterion; re-validate this relationship on current, real transactional data before acting on it.
Previous Purchases	More past purchases modestly raise predicted subscription likelihood, in line with general loyalty intuition.	Track repeat purchase count as an early signal for proactive loyalty/subscription offers.
Shipping Type	Faster/premium shipping choices (Next Day Air, 2-Day) trend slightly negative, while Express trends slightly positive — the effects are small and mixed.	Treat as a minor, secondary signal only; not a strong basis for standalone targeting.
Item Purchased / Size	A handful of specific product and size categories show small negative associations.	Monitor rather than act on — effect sizes are small relative to Discount Applied and Gender.

4. Important Caveat

Feature importance here reflects association, not proven causation. In particular, it is plausible that subscribing *causes* customers to receive more discounts (a membership perk) rather than discounts causing subscription — the direction of causality needs business/domain confirmation, not just model output. The strong, near-deterministic Gender effect should be treated as a data-quality finding to investigate, not a validated targeting rule.

5. Target Variable Note

The supplied dataset (ecommerce_customer_data.csv) contains no literal purchase/no-purchase outcome or browsing-session fields, since every row is already a completed transaction. Subscription Status (Yes/No) was used as the binary target as the closest genuinely binary, business-meaningful outcome available in the data, representing loyal/engaged customers versus one-off shoppers.